

SHUBHI SHRIVASTAV

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SUMMARY

Senior Software engineer with 5+ years of work experience in building scalable backend systems, cloud native microservices, and AI powered applications. Experienced in designing and deploying production LLM applications, RAG pipelines, AI agents, vector search systems and workflow orchestration platforms. Strong expertise in python, Kubernetes, docker, distributed systems, and cloud infrastructure with focus on building reliable and scalable enterprise AI solutions.

EDUCATION

University of Georgia, Franklin College of Arts & Sciences <i>Masters in Computer Science with specialisation in AI</i>	Athens, Georgia, US Jan 2021 - May 2022
University of Maryland, Baltimore County <i>(Coursework + research work in Information Systems)</i>	Baltimore, Maryland, US Jan 2020 - Dec 2020
PES University, Bengaluru <i>Bachelor of Technology, Computer Science</i>	Bangalore, India Aug 2014 - May 2018

WORK EXPERIENCE

Freelance Projects	Aug 2025 - Present
DocFlow — AI Document Processing Pipeline: Built a self-corrective agentic RAG system with LangGraph, hybrid search (dense, BM25, graph traversal), reciprocal rank fusion, and a custom evaluation harness. Engineered the full stack including semantic chunking, knowledge graph extraction, input/output guardrails, and real time streaming with FastAPI and Next.js. - Evaluated retrieval quality and response accuracy using RAGAS metrics (faithfulness, context precision); implemented guardrails for hallucination detection, PII filtering and input validation to ensure safe, compliant outputs. Atelier — Studio Management Platform: Building an AI platform for creative art studios to manage bookings and client interactions for workshops.	
Infinx Healthcare <i>AI Product Engineer</i>	remote, Bengaluru, India Jan 2025 - Aug 2025
- Led development of LLM-powered end to end application enabling real time autofill of health insurance forms for Aetna, Ambetter, Caredon etc streamlining processes which were previously handled by large internal support teams, reducing manual operational overload by around 40%, optimised performance by implementing frequency based learning and caching strategies. - Architected scalable ingestion pipelines using Scrapy and Selenium to extract and parse large scale healthcare policy guidelines by UHC, Medicare CMS level policies LCDs and NCDs, storing indexed content in MongoDB and S3 to power a production grade RAG system. - Built AI-driven agents including a voice assistant using streamlit, react and MCP for tool integration, and semantic search solutions integrating Azure OpenAI, whisper speech to text, vector databases, MindsDB and Neo4j. Implemented retrieval augmented querying over structured and unstructured data utilising vector search for accurate contextual responses. - Evaluated prompt strategies, retrieval quality and response accuracy for performance improvement	
NCR Corporation (NCR Voyix) <i>Software Engineer</i>	Atlanta, Georgia, US June 2022 - Sept 2024
- Lead design and development of scalable payments based microservices and APIs powering self checkout for major retailers like Walmart, Costco, Kroger, Publix etc improving customer success and reliability at scale - Containerized and deployed cloud native solutions on services using Kubernetes and Docker leveraging google cloud, reducing downtime by 20% and increasing overall service availability - Owned database performance and data pipelines, developing python scripts for cleanup and data ingestion, improving onboarding efficiency of new self checkout internal devices by 80% - Built an internal AI chatbot (ChatAssist) to automate employee support queries, significantly reducing employee response times and improving operational efficiency - Optimised transaction processing APIs, improving checkout transaction time by 30% through performance and load testing - Improved system security and compliance by owning to upgrade legacy services and resolving critical vulnerabilities (eg: aquasec), ensuring stable production environments	
Terry College of Business	Athens, Georgia, US

- Evaluated cloud infrastructure configurations with Google compute engine to identify cost-effective deployment strategies for a large scale data analysis workloads
- Designed and automated a data ingestion pipeline using python and Tweepy to collect and process real time Twitter data
- Implemented scheduled automation scripts for continuous analysis of information dissemination patterns by social media bots
- Assisted professor with assignment gradings, project team formations, promoting best programming practices and provided mentorship and hands-on training to undergrads on fundamentals of python and java.

Electronics for Imaging (EFI)

Bengaluru, India

Associate Software Engineer + Internship

Jan 2018 - Nov 2019

- Developed backend features using Java for industrial printing workflows (purchasing, scheduling, inventory, job costing), conducted regression, unit and integration testing
- Enhanced code quality by performing static code analysis by 89% using SonarQube and Findbugs tools
- Gained hands-on experience working with Spring framework, git, CI/CD pipelines and Agile development

PROJECTS

DocFlow — AI Workflow Automation Platform (Freelance)

- Self-corrective agentic RAG system with LangGraph, hybrid search (dense, BM25, graph traversal), reciprocal rank fusion, and custom evaluation harness. Full stack with semantic chunking, knowledge graph extraction, input/output guardrails, and real time streaming with FastAPI and Next.js.

Atelier — Studio Management Platform (Freelance)

- AI-powered self-service platform for creative art studios to manage bookings, client interactions, and workshop scheduling using LLM-based conversational agents.

Adversarial Robustness in Object Detection models in Self-driving Cars, *Trustworthy ML (master's project)*

- Served as a project team lead and analysed various malicious attack simulations on LIDAR object detection models algorithms used by autonomous driving vehicles

Online Bookstore website - forWords

- Collaborated with a team to design and build a full stack online bookstore website using Bootstrap, thyme leaf for integration and Java spring boot backend with micro services architecture, using MySQL as database

COVID contact tracing Backend (PL/SQL, Oracle)

- Built a relational backend system with PL/SQL procedures for data processing and validation and multi level queries for COVID related datasets for COVID contact tracing. Designed scalable schema and many-to-many relationships for efficient data modelling.

TECHNICAL SKILLS

Programming Languages: Java, Python, Golang, SQL

Backend & Systems: Microservices, Springboot, RESTful APIs, CI/CD pipelines

Databases: MySQL, Postgres, MongoDB, Vector databases (ChromaDB, FAISS, Qdrant), Redis

Cloud & Infra: AWS, GCP (GKE, Pub/Sub), Kubernetes

AI/ML: RAG, Agentic AI, Prompt engineering, Vector Search, AI agents, Langchain, Claude assisted development